

MKTG 6037 MA/MB

Machine Learning and AI in Marketing

COURSE INFORMATION

Instructor:	Prof. Jingbo Wang	Email:	jingbowang@cuhk.edu.hk
MKTG 6037MA:	Mon 13:30–16:30	Venue:	Cheng Yu Tung Bldg 410
MKTG 6037MB:	Mon 18:15–21:15	Venue:	Cheng Yu Tung Bldg 615
Teaching Assistant:	Mr. Yifeng Zeng	TA Email:	lukezeng@cuhk.edu.hk
Office Hours:	By appointment; please email to schedule		
Term Dates:	March 9 – June 1, 2026		
Course Site:	Blackboard (materials, announcements, and assignment submissions)		

COURSE DESCRIPTION

This practitioner-oriented course equips students with the knowledge and hands-on skills to harness deep learning and modern Generative AI for data-driven marketing decisions. Using the AI-powered coding environment *Trae*, students learn to collaborate with AI assistants deliberately and effectively—moving beyond passive prompt-and-hope usage toward strategic, context-aware interaction.

Through concise lectures, guided Python workshops, and hands-on labs, students design, build, and evaluate neural networks—Multilayer Perceptrons (MLPs), Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), and Transformers—applied to authentic marketing challenges including advertising-channel attribution, time-series forecasting, product-image classification, and customer-sentiment analysis. The course also demystifies how Large Language Models work under the hood, introduces agent-based AI development, guides students in deploying models as interactive web applications, explores responsible AI practices, and cultivates the professional mindset required of an AI-literate marketer.

LEARNING OUTCOMES

Upon successful completion of this course, students will be able to:

- Explain how Large Language Models function—from token prediction to the full training pipeline—and apply that understanding to interact with AI assistants effectively.
- Manage the AI context window and employ targeted selectors to provide the assistant with precisely the right information.
- Select the appropriate AI interaction mode (Agent, Ask, Edit) and configure persistent project-level rules to guide assistant behavior.
- Debug AI-generated code systematically, breaking the “generate–error–regenerate” cycle through structured diagnosis.
- Audit AI output critically—review diffs, reject unsound changes, and leverage version control as a safety net.
- Delegate multi-step development tasks to AI agents, applying structured oversight principles to manage autonomous code generation.
- Design, implement, and evaluate neural networks (MLP, RNN/LSTM, CNN, Transformer) for marketing applications across diverse data modalities.
- Justify the choice of deep-learning architecture for a given data type (tabular, sequential, image, text) and business context.
- Deploy trained models as interactive web applications and evaluate deployment platforms for sharing AI-powered marketing tools.
- Communicate model results, limitations, and ethical considerations in clear, professional, and accessible language.

PREREQUISITES

This course is designed to be largely self-contained and does not presuppose prior machine-learning experience. Familiarity with introductory calculus, linear algebra, and basic data-oriented programming in Python is beneficial; concise refreshers are provided as needed.

Expected Statistical Background: Intermediate

Expected Programming Proficiency: Advanced

TEXTBOOK & REFERENCES

- **Primary Reference:** Course lecture notes (provided; comprehensive and self-contained).
- Schwarz, N., Chapman, C., & Feit, E. M. (2020). *Python for Marketing Research and Analytics*. Springer.

GRADING AND EVALUATION

Assessment Components

The course grade is determined by the following components:

Assessment Component	Weight
Class engagement and attendance	20 %
Four applied assignments (10 % each)	40 %
Mid-term group presentation	20 %
Final take-home examination	20 %

Grading Descriptors (University Standard)

- A** Outstanding performance on all learning outcomes.
- A–** Generally outstanding performance on all (or almost all) learning outcomes.
- B** Substantial performance on all learning outcomes, or high performance on some outcomes that compensates for less satisfactory performance on others.
- C** Satisfactory performance on the majority of learning outcomes, possibly with a few weaknesses.
- D** Barely satisfactory performance on a number of learning outcomes.
- F** Unsatisfactory performance on a number of learning outcomes, or failure to meet specified assessment requirements.

COURSE POLICIES

Academic Integrity

Integrity is fundamental to the academic enterprise. All students are expected to uphold the highest standards of academic honesty in accordance with the University's *Policy on Academic Honesty* (https://www.cuhk.edu.hk/policy/academic_honesty). Plagiarism, unauthorized collaboration, and fabrication of data are serious offences that may result in a failing grade for the assignment or the course, and referral to the relevant disciplinary body. All written coursework must be submitted through **VeriGuide** with the signed honesty declaration attached.

Use of Generative AI

Approach: Encouraged with Full Attribution. In this course you are encouraged to use AI tools as a collaborative learning partner whenever permitted. All uses must be transparently acknowledged—including the tool name, model or version, date, and the key prompts employed. You remain solely responsible for the accuracy, originality, and ethical integrity of all submitted work.

Example attribution:

“I used ChatGPT (GPT-4o, 15 Jan 2026) to draft an initial outline for Section 2, using the prompt: ‘Summarize key challenges in marketing attribution for a graduate audience.’ I subsequently verified, restructured, and edited all content; all analysis, code, and citations are my own.”

Attendance and Participation

Regular attendance is expected and constitutes a meaningful component of the course grade. Students should complete assigned readings before each session and arrive prepared to contribute to discussions and in-class exercises. Participation credit cannot be made up retroactively. Absences supported by official documentation (e.g., medical certificate, university-approved activity) and communicated within **three business days** will be reviewed on a case-by-case basis.

Late Submission Policy

All assignments must be submitted by the stated deadline. Requests for extensions must be emailed to the instructor **before** the deadline and are granted at the instructor’s discretion. Without an approved extension, a penalty of **20 %** of the total available marks will be deducted for each 24-hour period (or part thereof) that the submission is late. Submissions more than **five calendar days** past the deadline will not be accepted and will receive a score of zero.

Collaboration and Individual Work

Each assignment will clearly specify whether individual or group work is expected. Unless an assignment is explicitly designated as collaborative, all submissions must represent the student’s own independent effort. Discussing concepts with peers is encouraged; sharing, copying, or paraphrasing code, written text, or solutions is prohibited. All external sources, tools, and collaborators must be credited appropriately.

Peer Evaluation for Group Work

For group-based components, peer evaluation will be conducted to ensure equitable contribution and discourage free-riding. Peer scores may adjust the relevant group component by up to 25 % of that task’s weight and by no more than 15 % of the total course grade. Evaluations will be administered through CATME (<https://catme.org>) or a designated peer-evaluation form. The instructor reserves the right to moderate scores in cases of clear anomalies; specific adjustment rules will be detailed on each assignment brief and reviewed in class.

Communication

Email is the primary channel for course-related correspondence. Please use your **CUHK email address** and include “MKTG 6037 MA/MB” in the subject line to facilitate prompt handling. The instructor will endeavor to respond within one business day during the teaching term. For substantive discussions regarding projects, academic concerns, or career matters, please schedule an office-hour appointment in advance.

Technology Requirements

This course uses **Python** (Anaconda distribution recommended) and the **Trae** AI IDE as its primary development environment. Students are responsible for maintaining a functional computing setup; detailed installation and configuration guides will be provided before the first assignment. Please bring a laptop to all sessions for in-class exercises unless otherwise instructed.

Use of Electronic Devices During Class

To foster a focused and respectful learning environment, electronic devices—including mobile phones, tablets, and laptops—should be used only for class-related activities during designated periods. Devices should be set to silent mode at all times. Persistent misuse may be reflected in the participation grade.

Data Ethics and Privacy

All datasets distributed for this course are to be used exclusively for course-related purposes. Students must not attempt to re-identify individuals, link datasets in ways that could compromise personal privacy, or distribute course data externally. Any sensitive or personally identifiable information must be removed before sharing code or outputs in public or semi-public forums (e.g., GitHub, discussion boards).

Inclusive Learning Environment

The Chinese University of Hong Kong is committed to fostering an inclusive, equitable, and respectful academic community. All members of this course—students, instructors, and teaching assistants—are expected to engage with one another in a manner that upholds dignity and mutual respect, regardless of background, identity, or viewpoint. Harassment, discriminatory language, or conduct that marginalizes others will not be tolerated and will be addressed in accordance with University policy. Students who have concerns are encouraged to speak with the instructor or to contact the appropriate University channels.

Recording Policy

Audio or video recording of any class session requires the **prior written consent** of the instructor and must comply with the University’s policies on intellectual property and privacy. Unauthorized recording or distribution of course materials is strictly prohibited.

Accessibility and Accommodations

Students who require academic accommodations due to a disability, chronic health condition, or other circumstance are encouraged to contact the *Office of Student Affairs* as early as possible and to inform the instructor so that appropriate arrangements can be made in a timely manner. The University and this course are committed to providing equitable access to learning for all students.

Student Well-Being

Your well-being matters. If you experience personal difficulties, health concerns, or excessive stress that may affect your academic performance, you are encouraged to reach out to the instructor or to the University's counseling and wellness services. Early communication enables us to explore reasonable accommodations and to connect you with the support you need.

Intellectual Property

All course materials—including lecture notes, slides, assignments, datasets, and code—are the intellectual property of the instructor and are provided for educational use within this course only. Reproduction, redistribution, or posting of these materials on external platforms without explicit written permission is prohibited.

CLASS SCHEDULE

The schedule below is provisional and subject to minor adjustments. Any changes will be announced in class and on the course site with reasonable advance notice.

MKTG 6037MA meets Mon 13:30 – 16:30 (CYT 410); MKTG 6037MB meets Mon 18:15 – 21:15 (CYT 615).

No class on Mar. 2 (Field Study Trip week). Final presentations on May 21 (CYT 615).

Week	Date & Topic	Content & Materials
1	Mar. 2, 2026 No Class – Field Study Trip	No class this week; reserved for Field Study Trip.
2	Mar. 9, 2026 Introduction, Logistics & Preparations	Materials: Lecture Notes, Sections 0–1 – The “Why”: from Passenger to Pilot when using AI coding assistants. – Course overview, learning objectives, readiness verification. – Why Python dominates AI; the ML library ecosystem; Conda environments. – Navigating the Trae IDE: editor panes, terminal, extensions, chat panel.
3	Mar. 16, 2026 How LLMs Work	Materials: Lecture Notes, Sections 2–3 – A brief history of LLMs: from Transformers (2017) to the Agent year (2026). – Embeddings, next-token prediction, and the training pipeline (pre-training, fine-tuning, RLHF). – Hallucinations, the stateless loop, and why LLMs fail. – The model landscape: choosing, controlling, and calling LLMs from Python. – Thinking models, multimodal models, and AI agents.
4	Mar. 23, 2026 Mastering Trae	Materials: Lecture Notes, Section 5 – The context window: the “paper towel tube” metaphor. – Three interaction modes: Agent, Ask, and Edit. – Precision targeting with # selectors; project-level rules. – Vibe debugging, code auditing, rollback, model selection, and exercises.

Week	Date & Topic	Content & Materials
5	Mar. 30, 2026 MLP: Which Ad Channel Drives Sales?	Materials: Lecture Notes, Section 6 – Business problem: multi-channel advertising attribution. – Data exploration, cleaning, and visualization. – Architecture: Multilayer Perceptron (neurons, layers, activation functions). – Vibe-coding an MLP with Trae; hyperparameter tuning; business interpretation.
6	Apr. 13, 2026 Welcome Back & Project Setup	<i>Assignment overview; form groups; review past presentation examples; draft and discuss final presentation proposals.</i>
7	Apr. 20, 2026 RNN: Can We Predict Stock Prices?	Materials: Lecture Notes, Section 7 – Business problem: time-series forecasting for marketing-relevant assets. – Data preprocessing: sliding windows, normalization, train/test splits. – Architecture: Recurrent Neural Networks and LSTMs (memory cells, gates). – Vibe-coding an LSTM with Trae; hyperparameter tuning; business interpretation.
8	Apr. 27, 2026 CNN: Sorting Products for an Online Store	Materials: Lecture Notes, Section 8 – Business problem: automated product image classification for e-commerce. – Image preprocessing: resizing, normalization, data augmentation. – Architecture: Convolutional Neural Networks (filters, pooling, feature maps). – Transfer learning: leveraging pre-trained models; business interpretation.
9	May 4, 2026 Transformers: What Are Customers Saying?	Materials: Lecture Notes, Section 9 – Business problem: customer sentiment analysis at scale. – Text preprocessing: tokenization, padding, vocabulary construction. – Architecture: Transformers and the attention mechanism. – Fine-tuning DistilBERT for sentiment; comparing approaches; business interpretation.

Week	Date & Topic	Content & Materials
10	May 11, 2026 Integration: The Right Tool for the Right Job	Materials: Lecture Notes, Section 10 – The architecture zoo: when to use MLP, RNN, CNN, or Transformers. – Evaluating models: choosing the right metric; the imbalance trap. – Responsible AI: bias, privacy, and transparency. – Capstone project briefing.
11	May 18, 2026 From Pilot to Product	Materials: Lecture Notes, Sections 11–13 – Agent-based development: from Pilot to Manager; Trae SOLO and Google Antigravity. – Building interactive dashboards with Gradio: turning trained models into web apps. – Deploying to the world: Render, Hugging Face Spaces, and Docker basics.
–	May 21, 2026 (Thu) Final Presentation Day (CYT 615)	MKTG 6037MA: 13 : 30 – 16 : 30 MKTG 6037MB: 18 : 15 – 21 : 15
12	Jun. 1, 2026 Take-Home Examination	<i>Take-home exam released and collected.</i>